Lecture 11: Node Voltage Analysis

EE201 – Circuit Theory I

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current voltage, node voltage e supernode

Review: two ways to solve an LTI resistive circuit

Transform the circuit

- Substitution, equivalent one-ports.
- Series and parallel connections.
- Source transformation.

Works for many circuits, but not all.

Keep the original circuit

- KCL: $A\mathbf{i} = \mathbf{0}$.
- KVL: $\mathbf{v} = A^T\mathbf{e}$.
- One element law per branch.

Works for every circuit.

With the node voltages **e** as the unknowns, KVL holds automatically.

Only KCL is left to write.

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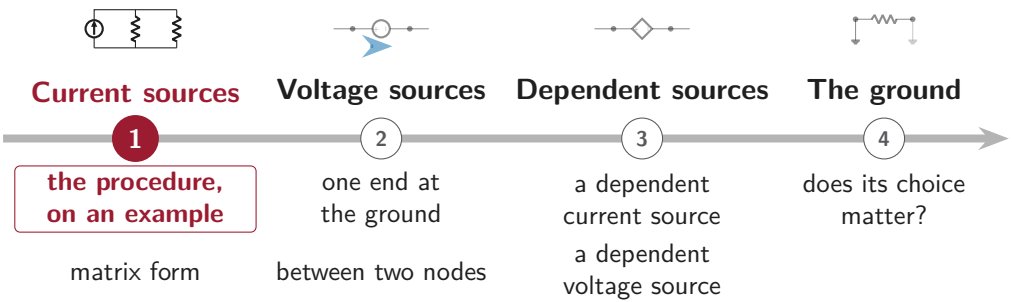
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Later, mesh current analysis will do the same with mesh currents.

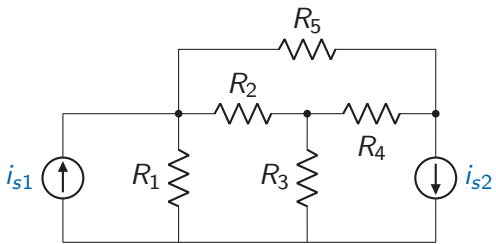
Roadmap



Each station adds a new kind of source; the last one asks about the ground.

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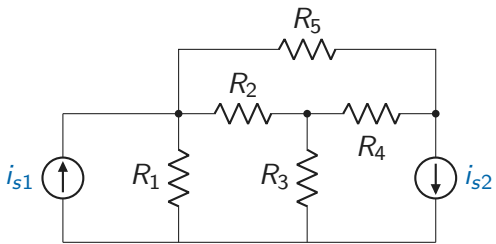
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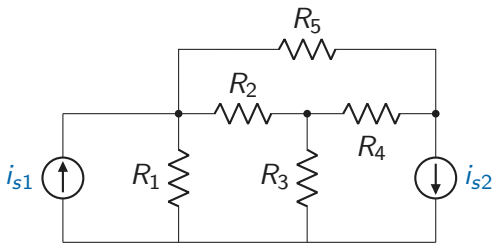


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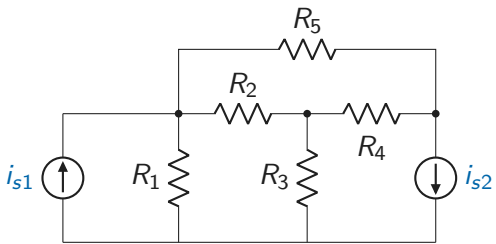
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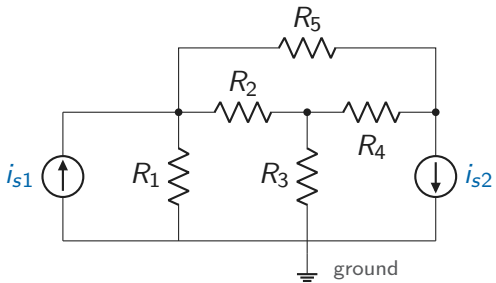
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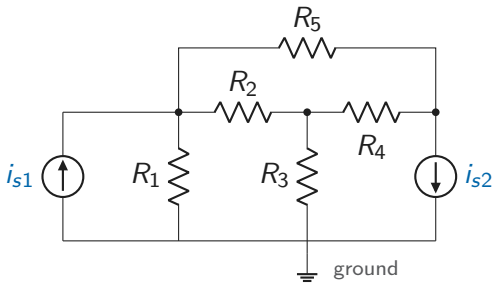
We find them in four steps, on this circuit.

Step 1: choose a reference node



Choose one node as the reference node, the ground.

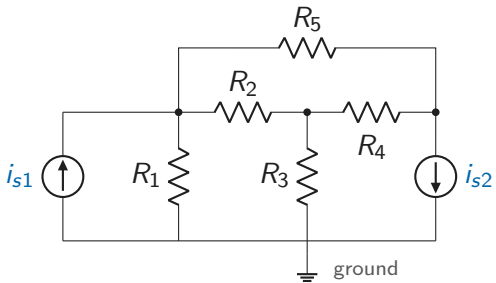
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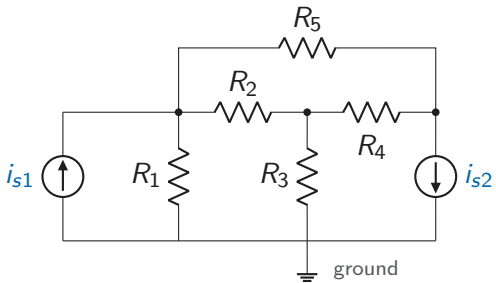


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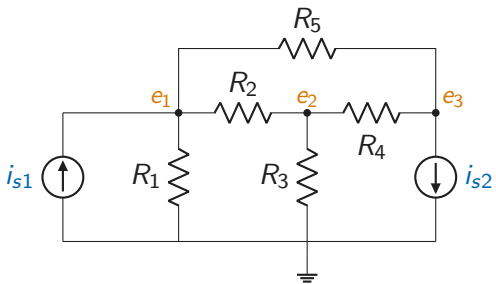
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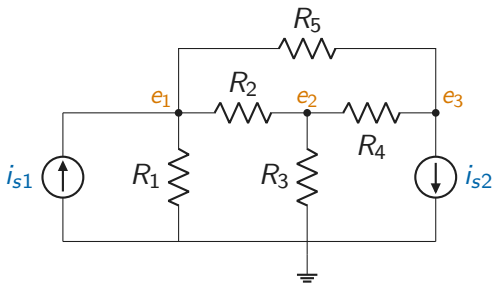
Any node can be the ground; we come back to this at the end.

Step 2: label the node voltages



Each other node gets a node voltage, measured from the ground: e_1, e_2, e_3.

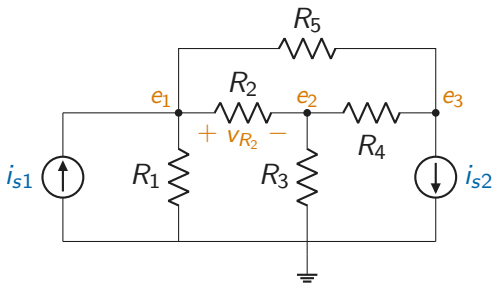
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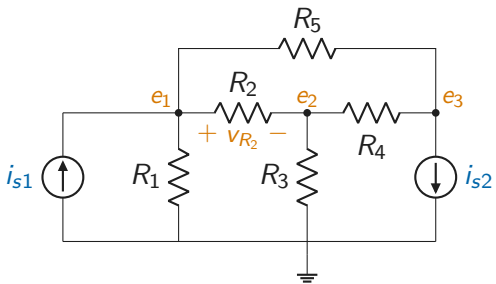


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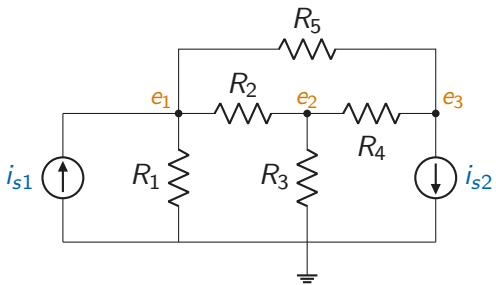
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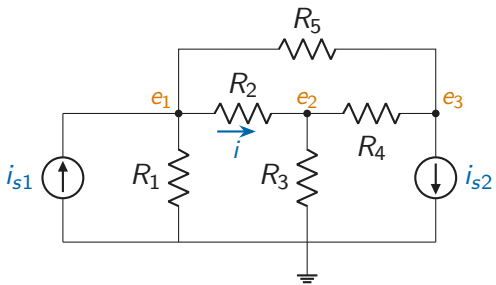
So KVL holds automatically: around R_1, R_2, R_3, $v_{R_1} = v_{R_2} + v_{R_3}$ is $e_1 = (e_1 - e_2) + e_2$.

Step 3: write the currents with node voltages



Ohm's law gives each resistor current from its two node voltages.

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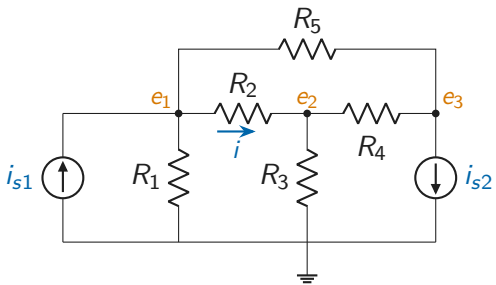


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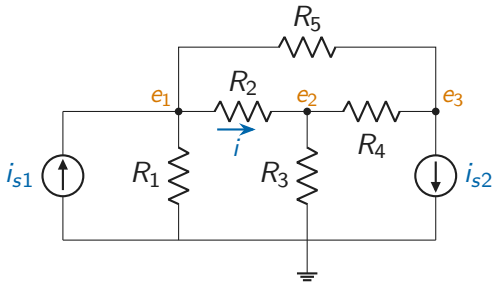
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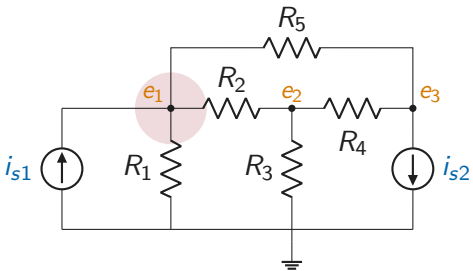
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The rule: own node voltage minus the other one, over R.

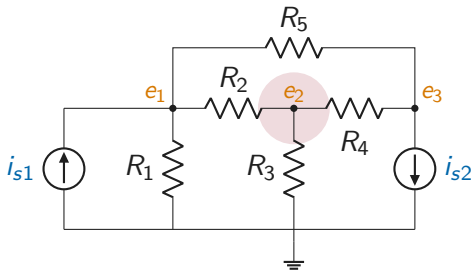
Step 4: KCL at every node except the ground



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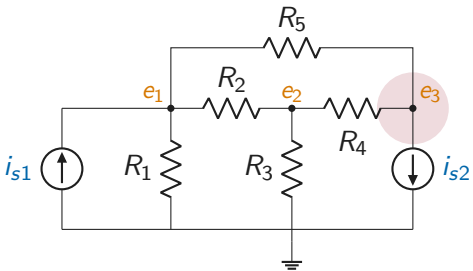
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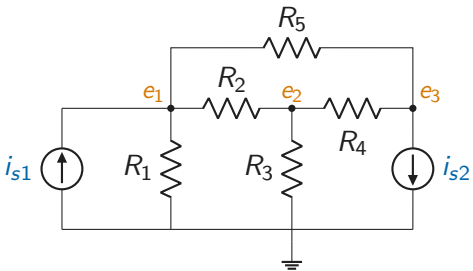
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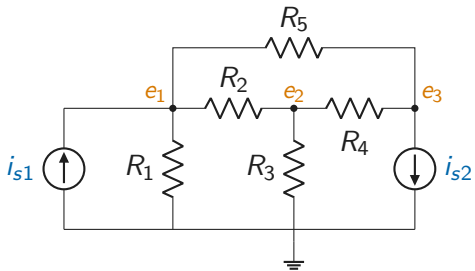
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A source pointing into the node enters with a minus sign.

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Three equations, three unknowns.

Then Steps 2 and 3 give every branch voltage and current.

Roadmap









Current sources | **Voltage sources** | **Dependent sources** | **The ground**

1 — 2 — 3 — 4

the procedure, on an example ✓

matrix form

one end at the ground

between two nodes

a dependent current source

a dependent voltage source

does its choice matter?

Each station adds a new kind of source; the last one asks about the ground.

Example 11.1 in matrix form

With the conductances $G_k = 1/R_k$:

$$\begin{aligned}
(G_1 + G_2 + G_5)\, e_1 - G_2\, e_2 - G_5\, e_3 &= i_{s1} \\
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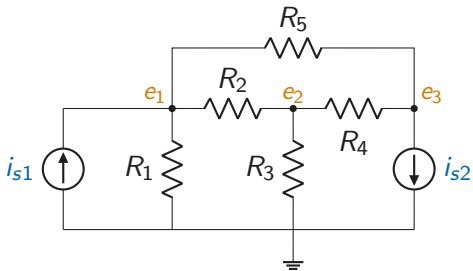
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Y_n is the node admittance matrix: $Y_n \mathbf{e} = \mathbf{i}_s$.

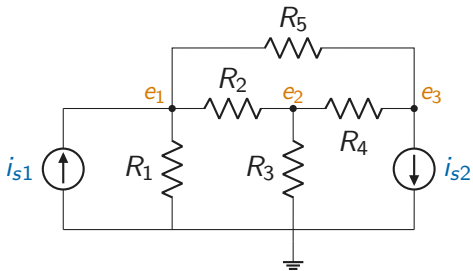
Writing Y_n directly from the circuit



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Question. Can you read each entry of Y_n directly from the circuit?

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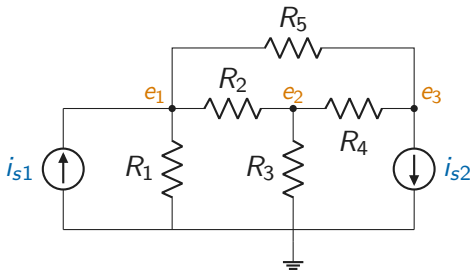


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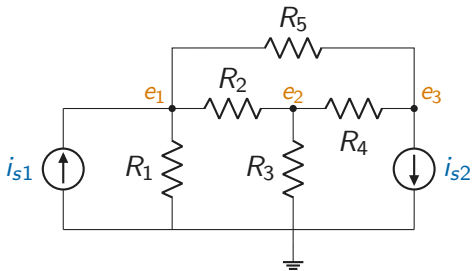
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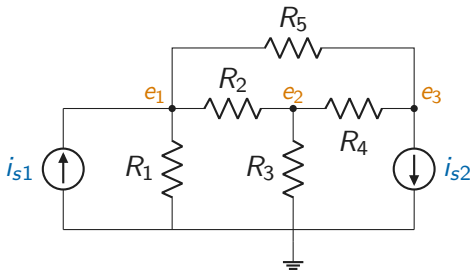
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Right side k: the sum of the source currents entering node k.

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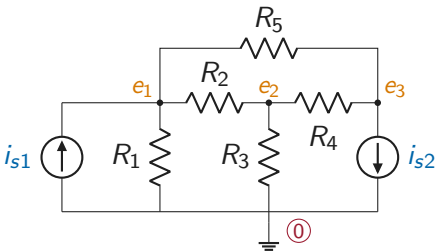
Diagonal (k,k): the sum of the conductances connected to node k.

Off-diagonal (k,j): minus the sum of the conductances between nodes k and j.

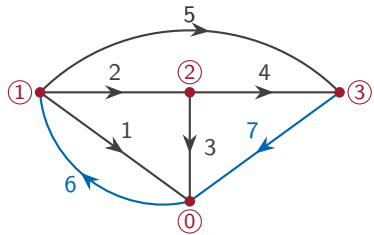
Right side k: the sum of the source currents entering node k.

So Y_n is symmetric: the entry (k,j) equals the entry (j,k).

Y_n from the graph: the reduced incidence matrix

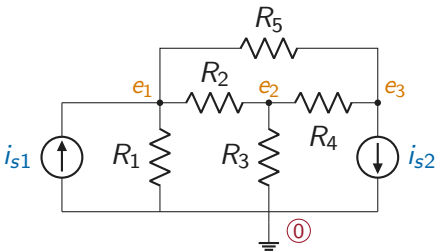


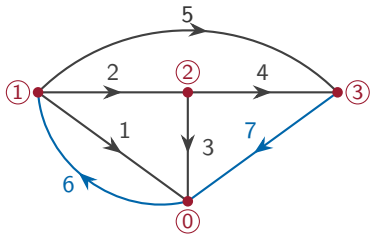
Question. Write the reduced incidence matrix A.



Branches 1–5: $R_1, \ldots, R_5$; 6, 7: i_{s1}, i_{s2}; node ⓪: the ground.

Y_n from the graph: the reduced incidence matrix





Branches 1–5: $R_1, \ldots, R_5$; 6, 7: i_{s1}, i_{s2}; node ⓪: the ground.

Question. Write the reduced incidence matrix A.

One row for each node except the ground.

Entry $+1$: the branch leaves the node; -1: it enters.

$$A = \begin{array}{c} \\ ① \\ ② \\ ③ \end{array} \overset{\begin{array}{ccccccc} 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{array}}{\left[\begin{array}{ccccc|cc} 1 & 1 & 0 & 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & -1 & 0 & 1 \end{array}\right]}$$

A_R: resistors A_S: sources

Y_n from the graph: the general result

- KCL: $A\mathbf{i} = \mathbf{0}$, that is $A_R\mathbf{i}_R + A_S\mathbf{i}_S = \mathbf{0}$.

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Put them together:

$$A_R\, G\, A_R^T\, \mathbf{e} = -A_S\mathbf{i}_S$$

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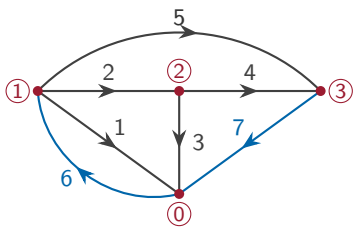
$$Y_n = A_R\,G\,A_R^T, \qquad \mathbf{i}_s = -A_S\mathbf{i}_S$$

$$Y_n^T = A_R\,G^T A_R^T = A_R\,G\,A_R^T = Y_n, \text{ since } G \text{ is diagonal.}$$

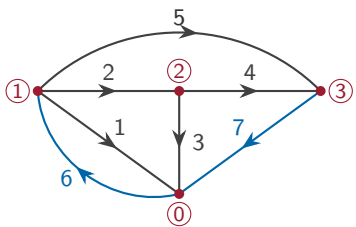
So Y_n is always symmetric for resistors and current sources.

Y_n from the graph: Example 11.1, the resistor voltages

A_R^T: the resistor columns 1–5 of A, written as rows.



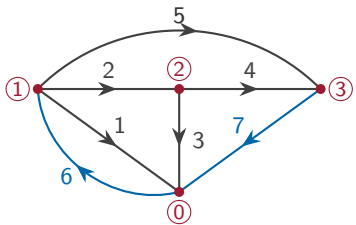
Y_n from the graph: Example 11.1, the resistor voltages



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$$\underbrace{\begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{bmatrix}}_{\mathbf{v}_R} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & -1 \\ 1 & 0 & -1 \end{bmatrix}}_{A_R^T} \underbrace{\begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix}}_{\mathbf{e}}$$

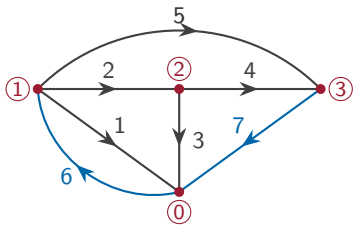
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Y_n from the graph: Example 11.1, the resistor voltages

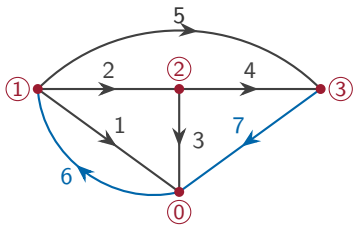


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Each resistor voltage: tail node voltage minus head node voltage.

Y_n from the graph: Example 11.1, the resistor voltages



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Each resistor voltage: tail node voltage minus head node voltage.

Then Ohm's law: $\mathbf{i}_R = G\,\mathbf{v}_R$, $i_k = G_k v_k$.

Y_n from the graph: Example 11.1, the matrix

$$A_R\, G = \begin{bmatrix} G_1 & G_2 & 0 & 0 & G_5 \\ 0 & -G_2 & G_3 & G_4 & 0 \\ 0 & 0 & 0 & -G_4 & -G_5 \end{bmatrix}$$

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$$A_R\, G\, A_R^T = \begin{bmatrix} G_1 + G_2 + G_5 & -G_2 & -G_5 \\ -G_2 & G_2 + G_3 + G_4 & -G_4 \\ -G_5 & -G_4 & G_4 + G_5 \end{bmatrix}$$

Y_n from the graph: Example 11.1, the matrix

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The same Y_n as from the three rules. ✓

Y_n from the graph: Example 11.1, the matrix

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$$-A_S \mathbf{i}_S = -\begin{bmatrix} -1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} i_{s1} \\ i_{s2} \end{bmatrix} = \begin{bmatrix} i_{s1} \\ 0 \\ -i_{s2} \end{bmatrix} = \mathbf{i}_s \quad ✓$$

Y_n from the graph: Example 11.1, the matrix

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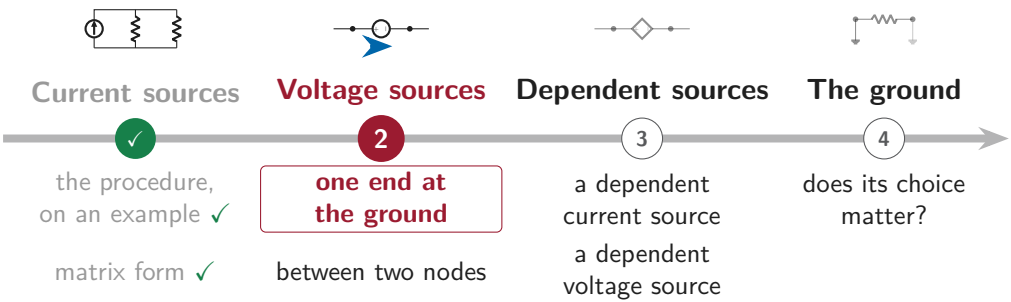
$$A_R\, G\, A_R^T = \begin{bmatrix} G_1 + G_2 + G_5 & -G_2 & -G_5 \\ -G_2 & G_2 + G_3 + G_4 & -G_4 \\ -G_5 & -G_4 & G_4 + G_5 \end{bmatrix}$$

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The rules are this product, read off entry by entry.

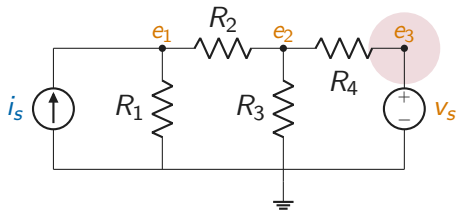
Roadmap



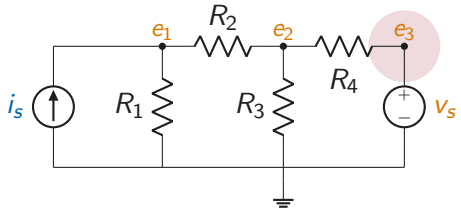
Each station adds a new kind of source; the last one asks about the ground.

Example 11.2: a voltage source to the ground

The source fixes node 3: $e_3 = v_s$ is known.



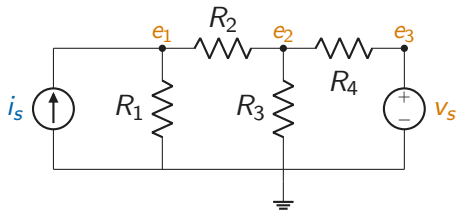
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Example 11.2: a voltage source to the ground



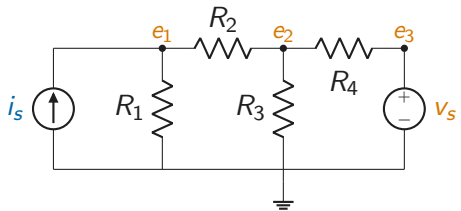
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Node 1: $-i_s + \dfrac{e_1}{R_1} + \dfrac{e_1 - e_2}{R_2} = 0$

Node 2: $\dfrac{e_2 - e_1}{R_2} + \dfrac{e_2}{R_3} + \dfrac{e_2 - v_s}{R_4} = 0$

Example 11.2: a voltage source to the ground



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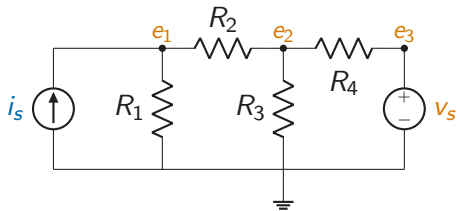
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$$\begin{bmatrix} \frac{1}{R_1} + \frac{1}{R_2} & -\frac{1}{R_2} \\ -\frac{1}{R_2} & \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} i_s \\ v_s/R_4 \end{bmatrix}$$

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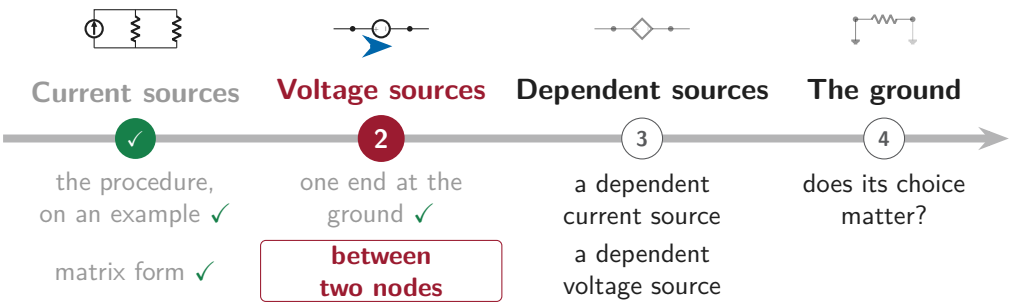
Node 1: $-i_s + \dfrac{e_1}{R_1} + \dfrac{e_1 - e_2}{R_2} = 0$

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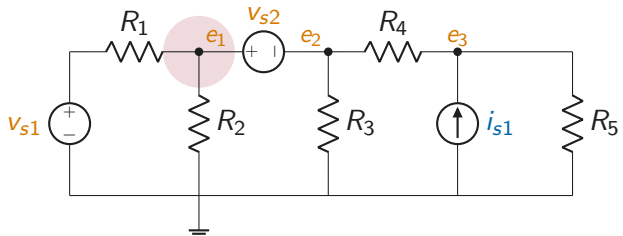
The known v_s moves to the right side.

Roadmap



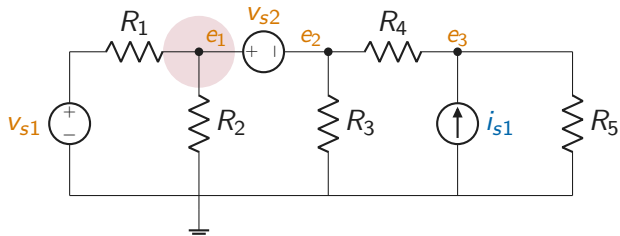
Each station adds a new kind of source; the last one asks about the ground.

Example 11.3: a voltage source between two nodes



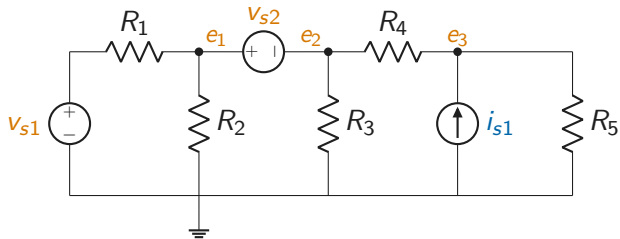
- **Question.** Write KCL at node 1: what is the current through v_{s2}?

Example 11.3: a voltage source between two nodes



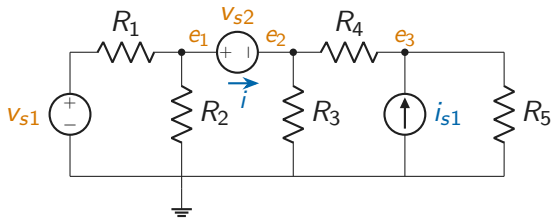
- **Question.** Write KCL at node 1: what is the current through v_{s2}?
- It cannot be written with e_1 and e_2: the law of a voltage source fixes its voltage, not its current.

Example 11.3: a voltage source between two nodes



- **Question.** Write KCL at node 1: what is the current through v_{s2}?
- It cannot be written with e_1 and e_2: the law of a voltage source fixes its voltage, not its current.
- **Modified node voltage analysis** offers two ways out: Method I (its current is a new unknown) and Method II (a supernode).

Method I: the current as an unknown



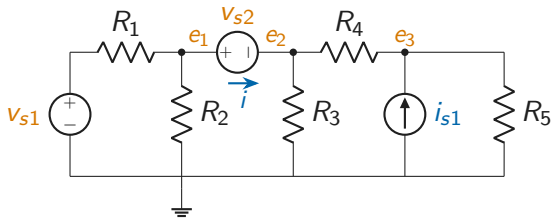
Node 1: $\dfrac{e_1 - v_{s1}}{R_1} + \dfrac{e_1}{R_2} + i = 0$

Node 2: $-i + \dfrac{e_2}{R_3} + \dfrac{e_2 - e_3}{R_4} = 0$

Node 3: $\dfrac{e_3 - e_2}{R_4} - i_{s1} + \dfrac{e_3}{R_5} = 0$

The current i of v_{s2} is one more unknown.

Method I: the current as an unknown



Node 1: $\dfrac{e_1 - v_{s1}}{R_1} + \dfrac{e_1}{R_2} + i = 0$

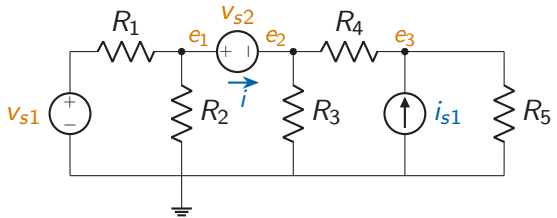
Node 2: $-i + \dfrac{e_2}{R_3} + \dfrac{e_2 - e_3}{R_4} = 0$

Node 3: $\dfrac{e_3 - e_2}{R_4} - i_{s1} + \dfrac{e_3}{R_5} = 0$

The current i of v_{s2} is one more unknown.

Constraint: $e_1 - e_2 = v_{s2}$.

Method I: the current as an unknown



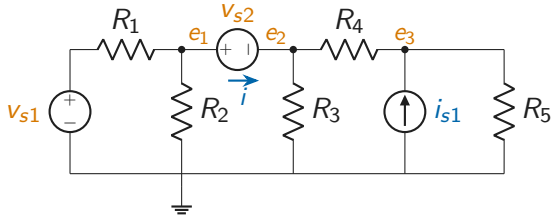
$$\begin{bmatrix} G_1+G_2 & 0 & 0 & 1 \\ 0 & G_3+G_4 & -G_4 & -1 \\ 0 & -G_4 & G_4+G_5 & 0 \\ 1 & -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ i \end{bmatrix} = \begin{bmatrix} G_1 v_{s1} \\ 0 \\ i_{s1} \\ v_{s2} \end{bmatrix}$$

The current i of v_{s2} is one more unknown.

Constraint: $e_1 - e_2 = v_{s2}$.

Four unknowns, four equations; the last row is the constraint.

Method I: the current as an unknown



$$\begin{bmatrix} G_1+G_2 & 0 & 0 & 1 \\ 0 & G_3+G_4 & -G_4 & -1 \\ 0 & -G_4 & G_4+G_5 & 0 \\ 1 & -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ i \end{bmatrix} = \begin{bmatrix} G_1 v_{s1} \\ 0 \\ i_{s1} \\ v_{s2} \end{bmatrix}$$

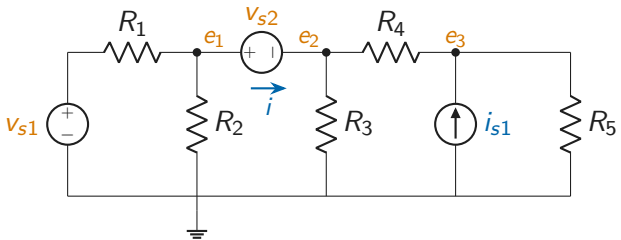
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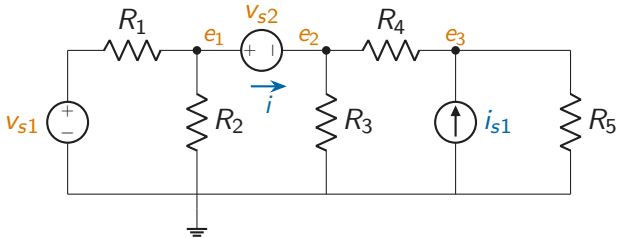
It works, but the supernode avoids the extra unknown.

Method II: the supernode



Question. Can we get rid of the unknown i?

Method II: the supernode

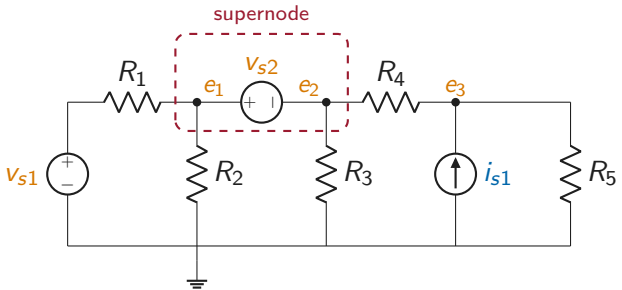


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i leaves node 1 and enters node 2, so it cancels.

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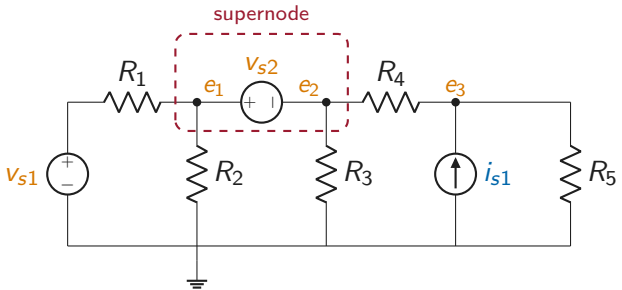
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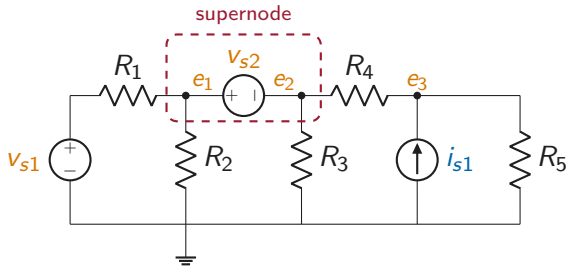
i leaves node 1 and enters node 2, so it cancels.

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Supernode

Write KCL as if the two nodes of the voltage source were a single node.

Method II: the equations



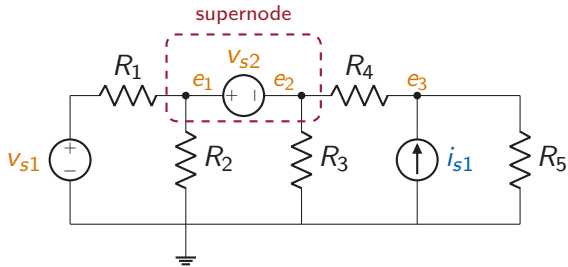
Supernode:

$$\frac{e_1 - v_{s1}}{R_1} + \frac{e_1}{R_2} + \frac{e_2}{R_3} + \frac{e_2 - e_3}{R_4} = 0$$

Node 3: $\dfrac{e_3 - e_2}{R_4} - i_{s1} + \dfrac{e_3}{R_5} = 0$

Constraint: $e_1 - e_2 = v_{s2}$

Method II: the equations



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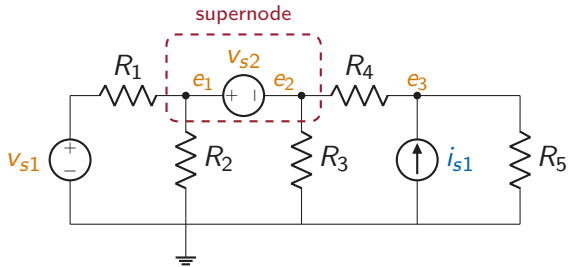
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$$\begin{bmatrix} G_1 + G_2 & G_3 + G_4 & -G_4 \\ 0 & -G_4 & G_4 + G_5 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} = \begin{bmatrix} G_1 v_{s1} \\ i_{s1} \\ v_{s2} \end{bmatrix}$$

Method II: the equations



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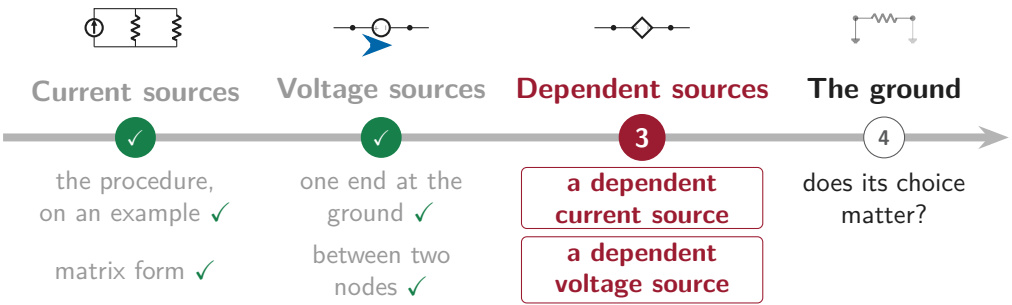
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Three unknowns: one fewer than Method I.

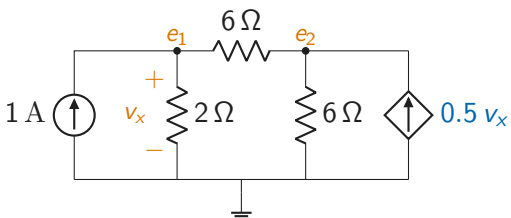
The matrix is not symmetric any more.

Roadmap



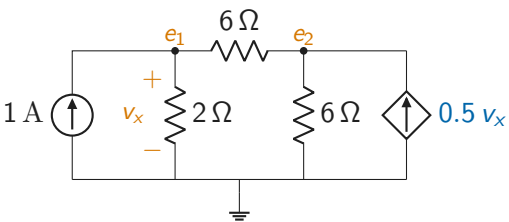
Each station adds a new kind of source; the last one asks about the ground.

Example 11.4: a dependent current source



Question. Find e_1 and e_2.

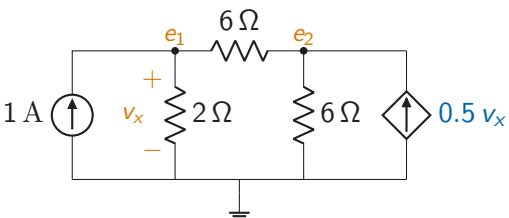
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Treat the dependent source as a current source of value $0.5\,v_x$.

Example 11.4: a dependent current source



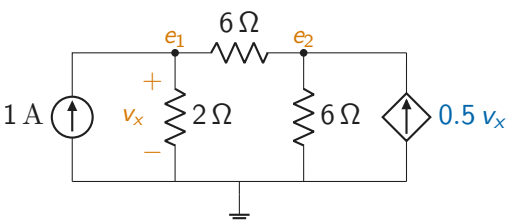
Question. Find e_1 and e_2.

Treat the dependent source as a current source of value $0.5\,v_x$.

Node 1: $$-1 + \frac{e_1}{2} + \frac{e_1 - e_2}{6} = 0$$

Node 2: $$\frac{e_2 - e_1}{6} + \frac{e_2}{6} - 0.5\,v_x = 0$$

Example 11.4: a dependent current source



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Treat the dependent source as a current source of value $0.5\,v_x$.

Node 1: $$-1 + \frac{e_1}{2} + \frac{e_1 - e_2}{6} = 0$$

Node 2: $$\frac{e_2 - e_1}{6} + \frac{e_2}{6} - 0.5\,v_x = 0$$

Write the control with node voltages: $v_x = e_1$.

Example 11.4: matrix form

As if $0.5\,v_x$ were an independent source:

$$\begin{bmatrix} \frac{2}{3} & -\frac{1}{6} \\ -\frac{1}{6} & \frac{1}{3} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0.5\,v_x \end{bmatrix}$$

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Put $v_x = e_1$ and move $0.5\,e_1$ to the left side:

$$\begin{bmatrix} \frac{2}{3} & -\frac{1}{6} \\ -\frac{1}{6} - \frac{1}{2} & \frac{1}{3} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \qquad \Rightarrow \qquad e_1 = 3\,\mathrm{V}, \quad e_2 = 6\,\mathrm{V}$$

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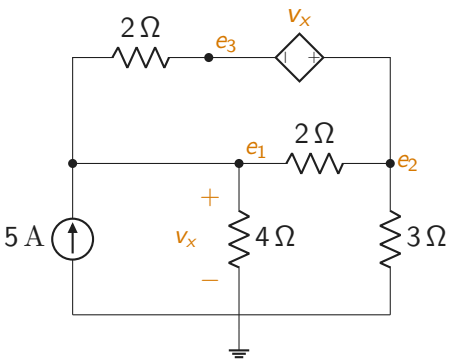
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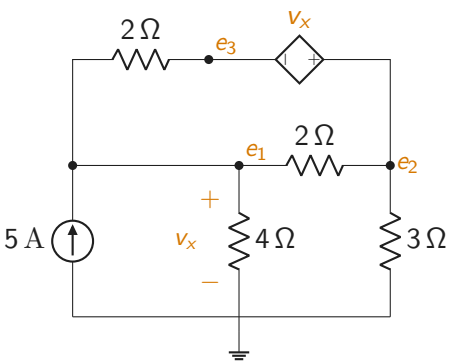
The dependent source changes one entry: Y_n is not symmetric any more.

Example 11.5: a dependent voltage source



Question. Find the power absorbed by the 3 Ω resistor.

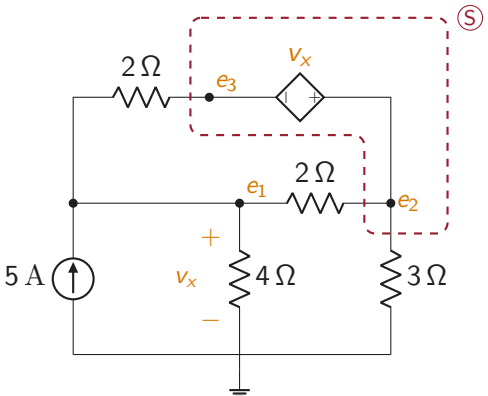
Example 11.5: a dependent voltage source



Question. Find the power absorbed by the 3 Ω resistor.

The voltage source v_x is between nodes 3 and 2; neither is the ground.

Example 11.5: a dependent voltage source

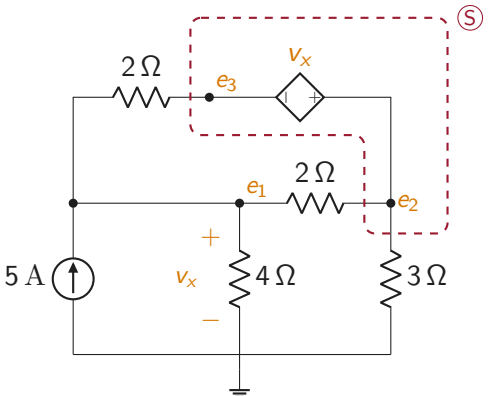


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So use a supernode Ⓢ around nodes 2 and 3.

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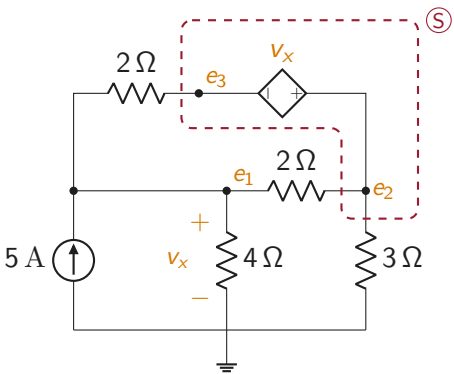
The voltage source v_x is between nodes 3 and 2; neither is the ground.

So use a supernode Ⓢ around nodes 2 and 3.

Unknowns: e_1, e_2, e_3.

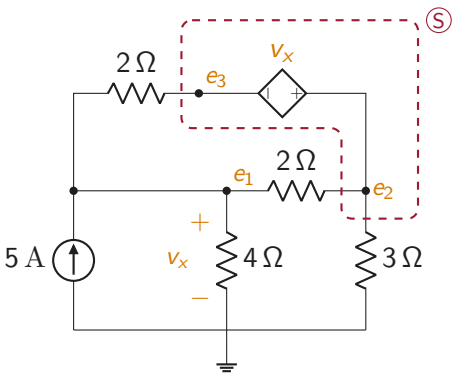
Equations: KCL at node 1, KCL at Ⓢ, and a constraint.

Example 11.5: the equations



Node 1: $$-5 + \frac{e_1 - e_3}{2} + \frac{e_1}{4} + \frac{e_1 - e_2}{2} = 0 \quad (1)$$

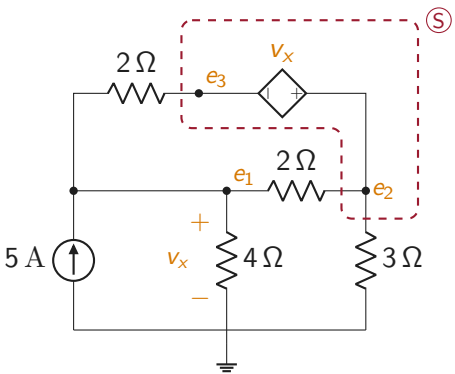
Example 11.5: the equations



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Example 11.5: the equations

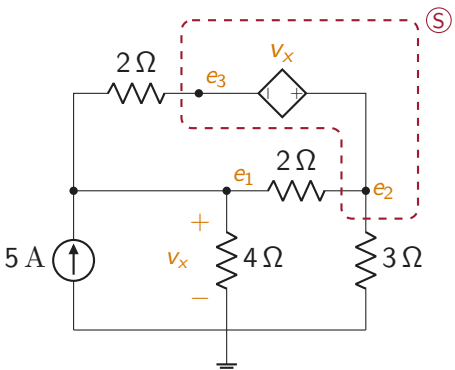


Node 1: $-5 + \dfrac{e_1 - e_3}{2} + \dfrac{e_1}{4} + \dfrac{e_1 - e_2}{2} = 0 \quad (1)$

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Question. What is the constraint?

Example 11.5: the equations



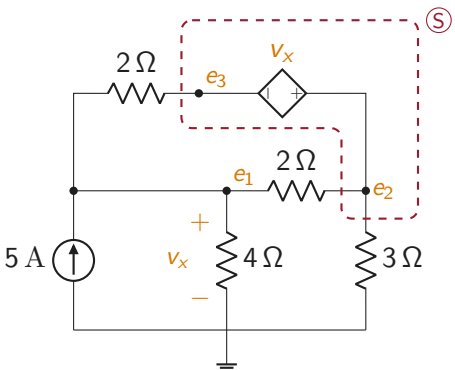
Node 1: $-5 + \dfrac{e_1 - e_3}{2} + \dfrac{e_1}{4} + \dfrac{e_1 - e_2}{2} = 0$ (1)

Supernode (S): $\dfrac{e_3 - e_1}{2} + \dfrac{e_2 - e_1}{2} + \dfrac{e_2}{3} = 0$ (2)

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The source: $e_2 - e_3 = v_x$.

Example 11.5: the equations



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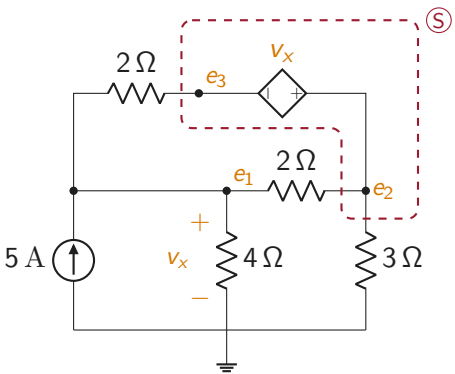
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The source: $e_2 - e_3 = v_x$.

Its control: $v_x = e_1$, the voltage of the 4 Ω.

Example 11.5: the equations



Node 1: $-5 + \dfrac{e_1 - e_3}{2} + \dfrac{e_1}{4} + \dfrac{e_1 - e_2}{2} = 0$ (1)

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Question. What is the constraint?

The source: $e_2 - e_3 = v_x$.

Its control: $v_x = e_1$, the voltage of the 4 Ω.

Constraint: $e_2 - e_3 = e_1$ (3)

Three equations, three unknowns.

Example 11.5: matrix form

Collect the terms of (1), (2) and (3):

$$\begin{bmatrix} \frac{5}{4} & -\frac{1}{2} & -\frac{1}{2} \\ -1 & \frac{5}{6} & \frac{1}{2} \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \\ 0 \end{bmatrix}$$

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Rows: KCL at node 1, KCL at Ⓢ, the constraint $e_1 - e_2 + e_3 = 0$.

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Solve: $e_1 = 8\,\mathrm{V}$, $e_2 = 9\,\mathrm{V}$, $e_3 = 1\,\mathrm{V}$.

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$$P_{3\Omega} = \frac{{e_2}^2}{3} = \frac{9^2}{3} = 27\,\mathrm{W}$$

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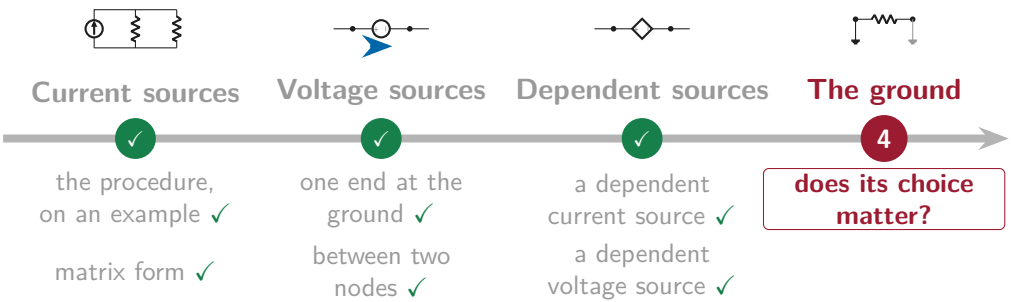
- A dependent source is handled like an independent one of the same kind.
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Remark

If a dependent voltage source sits between node k and the ground, do not write KCL at node k: its current is not known.

Write the constraint that the dependent source gives instead.

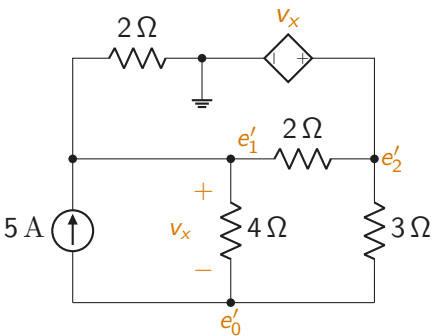
Roadmap



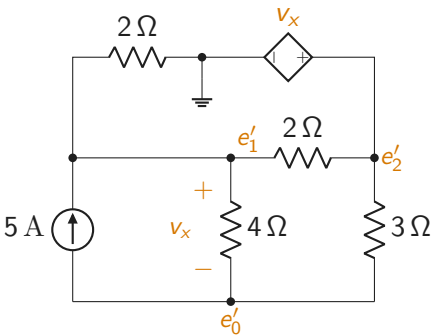
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Example 11.6: another ground

Question. Take node 3 of Example 11.5 as the ground. Does $P_{3\Omega}$ change?



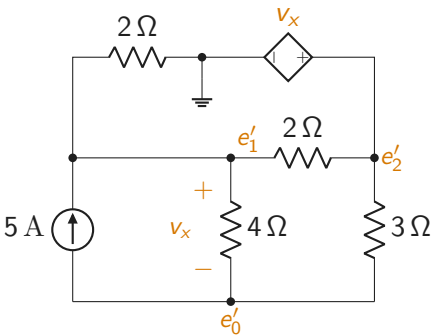
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Every node voltage drops by the same $e_3 = 1\,\mathrm{V}$:
$e_1' = 7\,\mathrm{V}$, $e_2' = 8\,\mathrm{V}$, $e_0' = -1\,\mathrm{V}$.

Example 11.6: another ground

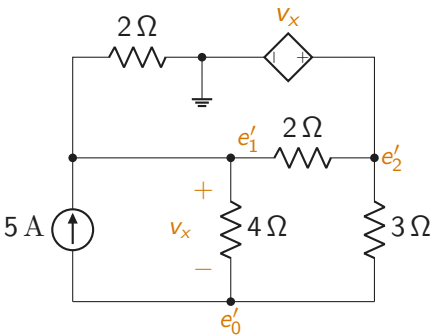


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$P_{3\Omega} = (e_2' - e_0')^2/3 = 27\,\mathrm{W}$, the same.

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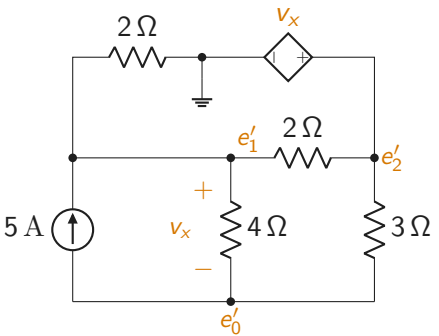
Now v_x is between node 2 and the ground: no supernode.

KCL at nodes 1, 0 and $e_2' = e_1' - e_0'$:

$$\begin{bmatrix} \frac{5}{4} & -\frac{1}{2} & -\frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{3} & \frac{7}{12} \\ 1 & -1 & -1 \end{bmatrix} \begin{bmatrix} e_1' \\ e_2' \\ e_0' \end{bmatrix} = \begin{bmatrix} 5 \\ -5 \\ 0 \end{bmatrix}$$

It gives the same 7, 8, $-1\,\mathrm{V}$.

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It gives the same 7, 8, $-1\,\mathrm{V}$.

Tip: put the ground at one end of a voltage source.

Summary

In the circuit	What to do
resistors and current sources	KCL at every non-ground node: $Y_n\mathbf{e} = \mathbf{i}_s$, from three simple rules; Y_n is symmetric
a voltage source from the ground to node k	$e_k = v_s$ is known; no KCL at node k
a voltage source between two nodes	Method I: its current as an unknown, plus the constraint Method II: KCL at the supernode, plus the constraint
a dependent source	write its control variable with node voltages: a constraint
the choice of ground	any node; the branch voltages and currents do not change

After the node voltages: $\mathbf{v} = A^T\mathbf{e}$, then the currents from the element laws.